M E S COLLEGE OF ARTS, COMMERCE AND SCIENCE

V Semester First Internal Test — Model Paper

III BCA Quantitative Techniques | Duration: 60 Minutes | Max Marks: 25

SECTION A

Answer any FIVE questions (2M × 5 = 10M)

Qn	Question Description	Marks
1	Check if the six-digit number 458,124 is completely divisible by 9. State the mathematical justification.	2M
2	Find the greatest number that will divide 148 and 246 leaving a remainder of 6 in each case.	2M
3	Find the remainder when 7^{19} is divided by 6 using the properties of congruence and remainders.	2M
4	If A's income is 25% more than B's, by what percentage is B's income less than A's? Apply shortcut or formula.	2M
5	A shopkeeper buys an article for Rs. 450 and sells it for Rs. 540. Calculate his profit percentage.	2M
6	An item with a Marked Price of Rs. 1,200 is offered with a discount of 15%. Find its Selling Price.	2M
7	If 15 men can complete a piece of work in 6 days, how many days will 10 men take to complete the same work under identical efficiency conditions?	2M

SECTION B

Answer any THREE questions (5M × 3 = 15M)

Qn	Question Description	Marks
8	A, B, and C start an enterprise by investing Rs. 20,000, Rs. 25,000, and Rs. 30,000 respectively. After 6 months, A withdraws Rs. 5,000 from the venture. If the total profit accumulated at the end of one year is Rs. 31,500, calculate B's share of the profit.	5M
9	A merchant marks his goods 40% above their cost price and then allows a discount of 20% on the marked price. Find his net profit percentage. If he originally purchased the goods for Rs. 1,500, calculate his actual profit amount in rupees.	5M
10	The product of two positive numbers is 1,575 and their ratio is 7 : 9. Find (i) the two numbers, and (ii) the Least Common Multiple (LCM) of these two numbers.	5M

11	In an examination, 40% of the students failed in English, 35% failed in Mathematics, and 15% failed in both subjects. If 400 students passed in both subjects, find the total number of students who appeared for the examination.	**5M**